

Colin Shea O'Brien

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Work Experience

SOFTWARE ENGINEER I/II – ROUTE APP

Jul 2020 - Dec 2022

System Design

- Designed and implemented internal authentication system and credential rotation process for SaaS product, reducing risk of revenue-impacting security vulnerabilities
- Enabled launch of new consumer web app with feature to associate shopping carts of unauthenticated users to authenticated ones by designing a flexible session-based token system
- Led the design, implementation, and migration from legacy billing system in SnapLogic to an improved serverless one, leveraging AWS Lambda, RDS, SQS, and SNS to provide scalability at negligible cost

Implementation

- Overhauled employee auth system by implementing SSO with OpenID Connect to facilitate secure access & eliminate the need for storing credentials
- Delivered new merchant auth system to address exposure of insecure auth tokens in browser storage
- Addressed unsolved memory leak, decreasing error count spikes in legacy monolith from 50k+ to <100
- Stood up ingestion platform to securely process 20 million customer emails per day without any PII leaks
- Ensured development of robust systems by building automated end-to-end testing framework
- Increased visibility into core services by implementing metrics, tracing, and logging, allowing for proactive responses more reliable systems, which especially contributed to stability over peak seasons like Black Friday

Mentorship & Training

- Led team and department level trainings on JWTs, TLS, mTLS, and other web authentication methods
- Mentored junior engineer on application development, testing processes, and system design, resulting in their ability to deliver billing service application and infrastructure
- Diagrammed & documented complex legacy system interactions, and trained team on ownership of several systems that had previously been obscured or ignored

ASSOCIATE SOFTWARE ENGINEER – IMAGINE LEARNING, INC.

Jan 2019 - Apr 2020

- Developed features and fixed bugs in test-driven process for educational Ruby on Rails application backend
- Built APIs for C# microservice, introduced Terraform to tech stack for configuring AWS services
- Integrated new applications into company framework and containerized them with Docker

WEB DEVELOPER – BRIGHAM YOUNG UNIVERSITY

Aug 2017 - Jan 2019

- Created full-stack role management web application in PHP/JavaScript, reducing processing time by 80%
- Improved user experience by finding and fixing over 200 web accessibility issues throughout website

Technical Skills

- **Expert:** GoLang, Terraform, AWS (ECS, Lambda, SQS, ElastiCache, etc.), Docker, GitLab CI
- **Proficient:** NodeJS, C#, MySQL, PostgreSQL, DynamoDB
- **Experienced:** ReactJS, Java, C++, C, Ruby on Rails, Python, Android

Education

Brigham Young University

Apr 2020

B.S. Computer Engineering, Minor in Computer Science